Modeling flux jumps in multiloop fluxonium

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I. MODEL A: JUMPS DUE TO DIRECT INCOHERENT TRANSITIONS

A. The system

The simplest model that possibly captures observed jumps in single-tone spectroscopy is a one-mode model for a fluxonium circuit with two linear inductors (labeled with A and B) forming two loops. Each loop has its own external flux Φ_{eA} and Φ_{eB}, and the fluxoid numbers are m_A and m_B respectively. The branch fluxes satisfy the sum rule constraints:

$$\Phi_{\mathrm{b}} + \Phi_{\mathrm{L,A}} = -\Phi_{\mathrm{e}A} - 2\pi m_A, \tag{1}$$

$$\Phi_{\mathrm{b}} + \Phi_{\mathrm{L,B}} = \Phi_{\mathrm{e}B} + 2\pi m_B. \tag{2}$$

Picking linear inductors as closure branches, the Hamiltonian is:

$$H = 4E_{\mathrm{C}}\hat{n}^2 + \frac{1}{2}[E_{\mathrm{LA}}(\hat{\phi} - \Phi_{\mathrm{e}A} - 2\pi\hat{m}_A)^2 + E_{\mathrm{LB}}(\hat{\phi} + \Phi_{\mathrm{e}B} + 2\pi\hat{m}_B)^2] - E_{\mathrm{J}}\cos\Big(\hat{\phi}\Big) \tag{3}$$

$$\begin{aligned} &= 4E_{\mathrm{C}}\hat{n}^2 + \frac{1}{2}E_{\mathrm{L}\Sigma}[\hat{\phi} - f_A(\Phi_{\mathrm{e}A} + 2\pi\hat{m}_A) + f_B(\Phi_{\mathrm{e}B} + 2\pi\hat{m}_B)]^2 - E_{\mathrm{J}}\cos\Big(\hat{\phi}\Big) \\ &\quad + \frac{1}{2}E_{\mathrm{L}\Sigma}\left\{f_A(\Phi_{\mathrm{e}A} + 2\pi\hat{m}_A)^2 + f_B(\Phi_{\mathrm{e}B} + 2\pi\hat{m}_B)^2 - [f_A(\Phi_{\mathrm{e}A} + 2\pi\hat{m}_A) - f_B(\Phi_{\mathrm{e}B} + 2\pi\hat{m}_B)]^2\right\} \end{aligned} \tag{4}$$

Here we define fractions $f_A = E_{\mathrm{LA}}/E_{\mathrm{L}\Sigma}$ and $f_B = E_{\mathrm{LB}}/E_{\mathrm{L}\Sigma}$. The energies E_{m_A} and E_{m_B} are the additional energies associated with the fluxoid numbers m_A and m_B. The Hilbert space of the system is $\mathcal{H} = \mathcal{H}_{\mathrm{flxn}} \otimes \mathcal{H}_A \otimes \mathcal{H}_B$, where $\mathcal{H}_A = \mathrm{Span}\{|m_A\rangle\}_{m_A}$ and $\mathcal{H}_B = \mathrm{Span}\{|m_B\rangle\}_{m_B}$ are the fluxoid number spaces for the two loops.

Define short-hand notations:

$$H = H_{\mathrm{smf}}[\hat{\Phi}_{\mathrm{eff}}(\hat{m}_A, \hat{m}_B)] + H_{\mathrm{offset}}(\hat{m}_A, \hat{m}_B) \tag{5}$$

$$H_{\mathrm{smf}}[\hat{\Phi}_{\mathrm{eff}}(\hat{m}_A, \hat{m}_B)] = 4E_{\mathrm{C}}\hat{n}^2 + \frac{1}{2}E_{\mathrm{L}\Sigma}(\hat{\phi} + \hat{\Phi}_{\mathrm{eff}})^2 - E_{\mathrm{J}}\cos\Big(\hat{\phi}\Big) \tag{6}$$

$$H_{\mathrm{offset}}(\hat{m}_A, \hat{m}_B) = \frac{1}{2}E_{\mathrm{L}\Sigma}f_A f_B(\Phi_{\mathrm{cm}} + 2\pi(\hat{m}_A + \hat{m}_B))^2 \tag{7}$$

$$\hat{\Phi}_{\mathrm{eff}} = -f_A(\Phi_{\mathrm{e}A} + 2\pi\hat{m}_A) + f_B(\Phi_{\mathrm{e}B} + 2\pi\hat{m}_B) \tag{8}$$

$$= 2\pi(-f_A\hat{m}_A + f_B\hat{m}_B) + \frac{-f_A + f_B}{2}\Phi_{\mathrm{cm}} + \frac{1}{2}\Phi_{\mathrm{diff}} \tag{9}$$

$$= 2\pi[-f_A(\hat{m}_A + \hat{m}_B) + \hat{m}_B] + \frac{-f_A + f_B}{2}\Phi_{\mathrm{cm}} + \frac{1}{2}\Phi_{\mathrm{diff}}. \tag{10}$$

An interesting observation is that, since the eigenenegies of the single-mode fluxonium Hamiltonian is invariant under the integer changes in $\hat{m}_B$, only when $\hat{m}_A + \hat{m}_B$ changes we would see changes in eigenenergies of H_{smf}. However, depending on the flux allocations, the effect of changing $\hat{m}_B$ on the eigenstates is non-trivial. The external flux dependence of the Hamiltonian: define common and differential fluxes Φ_{cm} and Φ_{diff} as:

$$\Phi_{\text{cm}} = \Phi_{\text{e}A} + \Phi_{\text{e}B}, \tag{11}$$

$$\Phi_{\text{diff}} = \Phi_{\text{e}A} - \Phi_{\text{e}B}, \tag{12}$$

and vice versa, expressing the external fluxes in terms of the common and differential fluxes:

$$\Phi_{\text{e}A} = \frac{1}{2}(\Phi_{\text{cm}} + \Phi_{\text{diff}}), \tag{13}$$

$$\Phi_{\text{e}B} = \frac{1}{2}(\Phi_{\text{cm}} - \Phi_{\text{diff}}). \tag{14}$$

The steps for simplifying the energy offset term $V(\hat{m}_A, \hat{m}_B)$ are:

$$H_{\text{offset}}(\hat{m}_A, \hat{m}_B) = \frac{1}{2} E_{\text{L}\Sigma} f_A f_B (\Phi_{\text{e}A} + 2\pi\hat{m}_A + \Phi_{\text{e}B} + 2\pi\hat{m}_B)^2 \tag{15}$$

$$= H_{\text{offset}}(\hat{m}_{\text{cm}}) \tag{16}$$

Define the common-mode fluxoid number $\hat{m}_{\text{cm}} = (\hat{m}_A + \hat{m}_B)$. The offset term H_{offset} only depends on $\hat{m}_{\text{cm}}$ and does not depend on $\hat{m}_A - \hat{m}_B$. We thus label the offset term as $H_{\text{offset}}(\hat{m}_{\text{cm}})$.

B. Symmetry properties of the Hamiltonian

Since $[\hat{m}_A, H] = [\hat{m}_B, H] = 0$, the fluxoid numbers are good quantum numbers for labeling eigenstates. To proceed on obtaining the full spectrum of the Hamiltonian, we thus diagonalize the Hamiltonian for each block given by each (m_A, m_B) sector. Once we fix a sector, the operators $\hat{m}_A$, $\hat{m}_B$ and $\hat{\Phi}_{\text{eff}}$ act as scalars:

$$\hat{m}_A \to m_A, \quad \hat{m}_B \to m_B, \quad \hat{\Phi}_{\text{eff}}(\hat{m}_A, \hat{m}_B) \to \hat{\Phi}_{\text{eff}}(m_A, m_B). \tag{17}$$

The Hamiltonian of this sector is thus:

$$H_{m'_A, m'_B} = H_{\text{smf}}[\Phi_{\text{eff}}(m'_A, m'_B)] + H_{\text{offset}}(m'_A + m'_B). \tag{18}$$

The eigenenergies and eigenvalues of $H_{\text{smf}}[\Phi_{\text{eff}}(m'_A, m'_B)]$ are labeled as $E_{\mu[\Phi_{\text{eff}}(m'_A,m'_B)]}$ and $|\mu[\Phi_{\text{eff}}(m'_A, m'_B)]\rangle$, respectively. Assembling these together, the full eigenenergies and eigenstates of $H_{m'_A,m'_B}$ are labeled as:

$$E_{\mu[\Phi_{\text{eff}}(m_A,m_B)],m_A,m_B} = E_{\mu[\Phi_{\text{eff}}(m_A,m_B)]} + H_{\text{offset}}(m_A + m_B) \tag{19}$$

$$|\mu[\Phi_{\text{eff}}(m_A, m_B)], m_A, m_B\rangle = |\mu[\Phi_{\text{eff}}(m_A, m_B)]\rangle \otimes |m_A\rangle \otimes |m_B\rangle\,. \tag{20}$$

Examining degeneracies in the spectrum. Let's compare the spectrum of each sector:

$$\mathcal{S}_{m_A,m_B} = \{(E_{\mu[\Phi_{\text{eff}}(m_A,m_B)],m_A,m_B}, |\mu[\Phi_{\text{eff}}(m_A, m_B)], m_A, m_B\rangle)\}_\mu. \tag{21}$$

if we change m_A by Δm_A, and m_B by Δm_B, the subspace Hamiltonian H_{m_A,m_B} changes as:

1. The effective external flux Φ_{eff} in the Hamiltonian $H_{m_A,m_B}[\Phi_{\text{eff}}]$ changes by $2\pi(-f_A\Delta m_A + f_B\Delta m_B)$.

2. The energy offset term $H_{\text{offset}}(m_A + m_B)$ changes as $H_{\text{offset}}(m_A + m_B) \to H_{\text{offset}}(m_A + m_B + \Delta m_A + \Delta m_B)$.

Specifically, under transformation with $\Delta m_A = k$ and $\Delta m_B = -k$,

$$\Phi_{\text{eff}} \to \Phi_{\text{eff}} - 2\pi k \tag{22}$$

$$H_{\text{offset}}(m_A + m_B) \to H_{\text{offset}}(m_A + m_B + k + -k) = H_{\text{offset}}(m_A + m_B) \tag{23}$$

$$H_{\text{smf}}[\Phi_{\text{eff}}] \to H_{\text{smf}}[\Phi_{\text{eff}} - 2\pi k] \tag{24}$$

$$\begin{aligned} H_{m_A,m_B} &= H_{\text{smf}}[\Phi_{\text{eff}}(m_A, m_B)] + H_{\text{offset}}(m_A + m_B) \\ &\to H_{m_A+k,m_B-k} = H_{\text{smf}}[\Phi_{\text{eff}}(m_A, m_B) - 2\pi k] + H_{\text{offset}}(m_A + m_B + k + -k). \end{aligned} \tag{25}$$

Since the eigenenergies of the H_{smf} part are periodic with respect to Φ_{eff} with period 2π, this implies that for any eigenstate with quantum numbers (m_A, m_B), there are infinitely many degenerate states labeled by $(m_A + k, m_B - k)$ for any integer k. Such shift can be understood as effectively adding k flux quanta to the loop A and subtracting k flux quanta from the loop B.

The $H_{\text{smf}}[\Phi_{\text{eff}}]$ part of the Hamiltonian has mirror symmetry around $\Phi_{\text{eff}} = 0$ and π. The eigenenergy spectrum is also periodic with period 2π. The energy offset part has mirror symmery around $\Phi_{\text{cm}} = -2\pi(m_A + m_B)$.

C. Jumps in the fluxoid numbers

Define operators $\hat{m}^+_{A,B} = \sum_{m_{A,B}} |m_{A,B}+1\rangle \langle m_{A,B}|$ and $\hat{m}^-_{A,B} = \sum_{m_{A,B}} |m_{A,B}-1\rangle \langle m_{A,B}|$ as raising and lowering operators for the fluxoid numbers. If the bath couples to the system via $\hat{m}^\pm_{A,B}$, we can then ask the decay rates between different eigenstates. Consider the system-bath coupling:

$$H_{\mathrm{SB}} = \hat{B}_A(\hat{m}^+_A + \hat{m}^-_A) + \hat{B}_B(\hat{m}^+_B + \hat{m}^-_B) \tag{26}$$

here $\hat{B}_A$ and $\hat{B}_B$ are the bath operators coupling to the raising and lowering operators of the fluxoid numbers. From here we can then compute the jump rates between different eigenstates by invoking Fermi's Golden Rule and assuming some noise spectral density. Consider these noise operator has respective spectral densities $S_A(E;T)$ and $S_B(E;T)$ and belongs to different baths, following FGR the transition rates between $|\mu[\Phi_{\mathrm{eff}}(m_A, m_B)], m_A, m_B\rangle$ (shorthand notation: $|\mu[m_A, m_B], m_A, m_B\rangle$) and $|\mu'[\Phi_{\mathrm{eff}}(m'_A, m'_B)], m'_A, m'_B\rangle$ (shorthand notation: $|\mu'[m'_A, m'_B], m'_A, m'_B\rangle$) are given by:

$$\begin{aligned}&\Gamma^A_{\mu[m_A,m_B],m_A,m_B\to\mu'[m'_A,m'_B],m'_A,m'_B} \\ &= \frac{1}{\hbar^2}|\langle\mu[m_A,m_B],m_A,m_B|\,\hat{m}^+_A + \hat{m}^-_A\,|\mu'[m'_A,m'_B],m'_A,m'_B\rangle|^2 S_A(E_{\mu[m_A,m_B]} - E_{\mu'[m'_A,m'_B]};T)\end{aligned} \tag{27}$$

$$\begin{aligned}&\Gamma^B_{\mu[m_A,m_B],m_A,m_B\to\mu'[m'_A,m'_B],m'_A,m'_B} \\ &= \frac{1}{\hbar^2}|\langle\mu[m_A,m_B],m_A,m_B|\,\hat{m}^+_B + \hat{m}^-_B\,|\mu'[m'_A,m'_B],m'_A,m'_B\rangle|^2 S_B(E_{\mu[m_A,m_B]} - E_{\mu'[m'_A,m'_B]};T).\end{aligned} \tag{28}$$

Focus on the $\hat{m}^+_A$ and $\hat{m}^-_A$ terms, the other terms are similar. The matrix element

$$\begin{aligned}&\langle n_F[\Phi_{\mathrm{eff}}(m_A,m_B)],m_A,m_B|\,\hat{m}^+_A\,|n'_F[\Phi_{\mathrm{eff}}(m'_A,m'_B)],m'_A,m'_B\rangle \\ &= \delta_{m_A,m'_A-1}\delta_{m_B,m'_B}\langle n'_F[\Phi_{\mathrm{eff}}(m'_A,m'_B)]|n_F[\Phi_{\mathrm{eff}}(m_A,m_B)]\rangle\end{aligned} \tag{29}$$

$$\begin{aligned}&\langle n_F[\Phi_{\mathrm{eff}}(m_A,m_B)],m_A,m_B|\,\hat{m}^-_A\,|n'_F[\Phi_{\mathrm{eff}}(m'_A,m'_B)],m'_A,m'_B\rangle \\ &= \delta_{m_A,m'_A+1}\delta_{m_B,m'_B}\langle n'_F[\Phi_{\mathrm{eff}}(m'_A,m'_B)]|n_F[\Phi_{\mathrm{eff}}(m_A,m_B)]\rangle.\end{aligned} \tag{30}$$

i.e., only jumps between nearest neighbor fluxoid numbers are allowed.

Notice that albeit the operators $\hat{m}^{+/-}_{A/B}$ act on the fluxoid number subspace, when acting onto the state it also changes the effective flux in the single-mode fluxonium part of the Hamiltonian.

Without assuming anything about the noise spectral density, the only thing we know is that the jump rates between a pair of eigenstates satisfy the detailed balance condition:

$$
\begin{aligned}
r_{\mu[m_A,m_B],m_A,m_B|\mu'[m_A,m_B],m_A+1,m_B} &= \frac{\Gamma_{\mu[m_A,m_B],m_A,m_B\to\mu'[m_A,m_B],m_A+1,m_B}}{\Gamma_{\mu'[m_A,m_B],m_A,m_B\to\mu[m_A,m_B],m_A+1,m_B}} \\
&= \exp\left(-\frac{E_{\mu'[m_A,m_B],m_A,m_B}-E_{\mu[m_A,m_B],m_A,m_B}}{k_BT}\right). \qquad (31)
\end{aligned}
$$

Given that the jump rates are in level of Hz and smaller, which is much smaller than the typical depolarization rates between eigenstates in a single-loop fluxonium, it is a good approximation to consider that after the changes in the fluxoid numbers, the system quickly thermalizes within the states that has the same fluxoid number:

$$
p_{\mu[m_A,m_B],m_A,m_B}(T) = \frac{\exp\left(-\frac{E_{\mu[m_A,m_B],m_A,m_B}}{k_BT}\right)}{Z_{m_A,m_B}}, \qquad (32)
$$

with the partition function

$$
Z_{m_A,m_B} = \sum_{\mu'} \exp\left(-\frac{E_{\mu'[m_A,m_B],m_A,m_B}}{k_BT}\right). \qquad (33)
$$

The total jump rate due to the noise acting on the loop A fluxoid number is then given by:

$$
\Gamma^A_{m_A,m_B\to m_A+1,m_B} = \sum_{\mu'} p_{\mu'[m_A,m_B],m_A,m_B} \times \Gamma^A_{\mu[m_A,m_B],m_A,m_B\to\mu'[m_A,m_B],m_A+1,m_B} \qquad (34)
$$

II. MODEL B: COHERENT-PHASE-SLIP-ASSISTED INCOHERENT JUMPS

A. The system

The setting is similar to the Model A, except that the states within different fluxoid/winding number sectors are now coupled, and the system couples to the bath via operators in the fluxonium subspace. An effective model for the double-loop fluxonium with the effect of coherent quantum phase slip is given by the following Hamiltonian:

$$
H_{\mathrm{S}} = H_{\mathrm{smf}} + H_{\mathrm{offset}} + H_{\mathrm{tun}}, \qquad (35)
$$

$$
H_{\mathrm{smf}} = 4E_{\mathrm{C}}\hat{n}^2 + \frac{1}{2}E_{\mathrm{L}\Sigma}(\hat{\phi}+\hat{\Phi}_{\mathrm{eff}})^2 - E_{\mathrm{J}}\cos(\hat{\phi}), \qquad (36)
$$

$$
\hat{\Phi}_{\mathrm{eff}} = -2\pi f_A\hat{P} + \frac{f_B-f_A}{2}\Phi_{\mathrm{cm}} + \frac{1}{2}\Phi_{\mathrm{diff}}, \qquad (37)
$$

$$
H_{\mathrm{offset}} = \frac{1}{2}E_{\mathrm{L}\Sigma}f_Af_B(\Phi_{\mathrm{cm}}+2\pi\hat{P})^2, \qquad (38)
$$

$$
H_{\mathrm{tun}} = E_{\mathrm{S}}\hat{I}_{\mathrm{smf}} \otimes \sum_P |P+1\rangle\langle P| + \mathrm{h.c.}. \qquad (39)
$$

Here, $E_{\mathrm{L}\Sigma} = E_{\mathrm{L}A} + E_{\mathrm{L}B}$, and $f_A = E_{\mathrm{L}A}/E_{\mathrm{L}\Sigma}$ and $f_B = E_{\mathrm{L}B}/E_{\mathrm{L}\Sigma}$ are the inductive energy ratios of the two inductors. The common and differential fluxes of the two loops are defined as $\Phi_{\mathrm{cm}} = \Phi_{\mathrm{e}A} + \Phi_{\mathrm{e}B}$ and $\Phi_{\mathrm{diff}} = \Phi_{\mathrm{e}A} - \Phi_{\mathrm{e}B}$ respectively. The Hilbert space of the system is $\mathcal{H} = \mathcal{H}_{\mathrm{smf}} \otimes \mathcal{H}_P$, where $\mathcal{H}_{\mathrm{smf}}$ is the Hilbert space of the single-mode fluxonium, and $\mathcal{H}_P = \mathrm{Span}\{|P\rangle\}_{P\in\mathbb{Z}}$ is the Hilbert space for the phase slip. Notice that here we use P to label the total number of phase slips along the two junction arrays, rather than using the winding number m_A and m_B counting for fluxoid numbers in each loop.

The tunneling strength E_{S} is given by the WKB/instanton calculation, which depends on the array junction parameters and offset charges on the junction array islands [1–3]:

$$E_{\mathrm{S}} = \sum_j E_{\mathrm{S},j}, \tag{40}$$

$$E_{\mathrm{S},j} = e^{2\pi i \eta_{\mathrm{g},j}} \times 2\sqrt{\frac{2}{\pi}}\sqrt{8E_{\mathrm{J}j}E_{\mathrm{C}j}}\left(\frac{8E_{\mathrm{J}j}}{E_{\mathrm{C}j}}\right)^{1/4}\exp\left(-\sqrt{\frac{8E_{\mathrm{J}j}}{E_{\mathrm{C}j}}}\right). \tag{41}$$

Here, $\eta_{\mathrm{g},j}$ is the sum of offset charges of junction islands up to j:

$$\eta_{\mathrm{g},j} = \sum_{k=1}^{j} n_{\mathrm{g},k}. \tag{42}$$

We label the eigenstates of the single-mode fluxonium Hamiltonian $H_{\mathrm{smf}}[\Phi_{\mathrm{eff}}(P)]$ as $|\mu[\Phi_{\mathrm{eff}}(P)]\rangle$, with eigenvalues $E_{\mu[\Phi_{\mathrm{eff}}(P)]}$. Treating the H_{tun} as a perturbation, denoting the rest Hamiltonian as H_0, the bare eigenstates and eigenenergies of the full Hamiltonian are given by:

$$E_{\mu[\Phi_{\mathrm{eff}}(P)],P} = E_{\mu[\Phi_{\mathrm{eff}}(P)]} + E_{\mathrm{offset}}(P), \tag{43}$$

$$|\mu[\Phi_{\mathrm{eff}}(P)],P\rangle = |\mu[\Phi_{\mathrm{eff}}(P)]\rangle \otimes |P\rangle. \tag{44}$$

The standard perturbation theory gives the dressed eigenenergies and eigenstates as:

$$\left|\overline{\mu[\Phi_{\mathrm{eff}}(P)],P}\right\rangle = |\mu[\Phi_{\mathrm{eff}}(P)],P\rangle + \sum_{P'} \frac{\langle\mu[\Phi_{\mathrm{eff}}(P)]|\, H_{\mathrm{tun}}\, |P'\rangle}{E_{\mu[\Phi_{\mathrm{eff}}(P)]} - E_{P'}} |\mu[\Phi_{\mathrm{eff}}(P')],P'\rangle. \tag{45}$$

The standard perturbation theory gives the dressed eigenenergies and eigenstates as:

a. Matrix elements of the tunneling Hamiltonian. The tunneling Hamiltonian couples adjacent P sectors:

$$\langle\nu[\Phi_{\mathrm{eff}}(P')],P'|H_{\mathrm{tun}}|\mu[\Phi_{\mathrm{eff}}(P)],P\rangle \tag{46}$$

$$= E_{\mathrm{S}}\delta_{P,P'+1}\langle\nu[\Phi_{\mathrm{eff}}(P')]|\mu[\Phi_{\mathrm{eff}}(P)]\rangle. \tag{47}$$

We define the matrix element

$$T_{\nu[P']\mu[P]} \equiv \langle \nu[\Phi_{\text{eff}}(P')]|\mu[\Phi_{\text{eff}}(P)]\rangle \, . \tag{48}$$

b. First-order energy correction. Since H_{tun} only connects P to $P \pm 1$, the diagonal matrix elements vanish:

$$E^{(1)}_{\mu[P],P} = 0. \tag{49}$$

c. Second-order energy correction. The leading correction to the energy is second order:

$$E^{(2)}_{\mu[P],P} = \sum_{\nu} \frac{E_{\text{S}}^2 \left|T_{\nu[P+1]\mu[P]}\right|^2}{E_{\mu[P],P} - E_{\nu[P+1],P+1}} + \sum_{\nu} \frac{E_{\text{S}}^2 \left|T_{\mu[P-1]\nu[P]}\right|^2}{E_{\mu[P],P} - E_{\nu[P-1],P-1}}. \tag{50}$$

Thus the dressed eigenenergies are

$$\overline{E}_{\mu[P],P} = E_{\mu[P],P} + \sum_{\nu} \left[\frac{E_{\text{S}}^2 \left|T_{\nu[P+1]\mu[P]}\right|^2}{E_{\mu[P],P} - E_{\nu[P+1],P+1}} + \frac{E_{\text{S}}^2 \left|T_{\mu[P-1]\nu[P]}\right|^2}{E_{\mu[P],P} - E_{\nu[P-1],P-1}} \right]. \tag{51}$$

d. First-order correction to eigenstates. The dressed eigenstates are given by

$$\begin{aligned} \left|\overline{\mu[P],P}\right\rangle = \left|\mu[P],P\right\rangle &+ \sum_{\nu} \frac{E_{\text{S}} T_{\nu[P+1]\mu[P]}}{E_{\mu[P],P} - E_{\nu[P+1],P+1}} \left|\nu[P+1],P+1\right\rangle \\ &+ \sum_{\nu} \frac{E_{\text{S}} T_{\mu[P-1]\nu[P]}}{E_{\mu[P],P} - E_{\nu[P-1],P-1}} \left|\nu[P-1],P-1\right\rangle . \end{aligned} \tag{52}$$

e. Near-degenerate case. When $E_{\mu[P],P} \approx E_{\nu[P+1],P+1}$, perturbation theory breaks down and one must diagonalize the effective Hamiltonian in the subspace $\{\left|\mu[P],P\right\rangle, \left|\nu[P+1],P+1\right\rangle\}$:

$$H_{\text{eff}} = \begin{pmatrix} E_{\mu[P],P} & E_{\text{S}} T_{\nu[P+1]\mu[P]} \\ (E_{\text{S}} T_{\nu[P+1]\mu[P]})^* & E_{\nu[P+1],P+1} \end{pmatrix}. \tag{53}$$

The resulting eigenenergies are

$$\overline{E}_{\pm} = \frac{E_{\mu[P],P} + E_{\nu[P+1],P+1}}{2} \pm \sqrt{\left(\frac{\Delta}{2}\right)^2 + E_{\text{S}}^2 \left|T_{\nu[P+1]\mu[P]}\right|^2}, \tag{54}$$

where $\Delta = E_{\mu[P],P} - E_{\nu[P+1],P+1}$. Clearly, the size of the avoided crossing is determined by the tunneling strength E_{S}.

B. Indirect flux jump due to bath coupling to fluxonium subsystem

Consider the system-bath coupling of the following form:

$$H_{\mathrm{SB}} = \sum_{\alpha} \hat{O}_{\mathrm{smf},\alpha} \hat{B}_{\alpha}, \tag{55}$$

where $\hat{O}_{\mathrm{smf},\alpha}$ is the operator of the fluxonium subsystem, and $\hat{B}_{\alpha}$ is the corresponding bath operator. Some typical examples are:

- Capacitive/dielectric loss: $\hat{O}_{\mathrm{smf}} = \hat{n}$
- Inductive loss: $\hat{O}_{\mathrm{smf}} = \hat{\phi}$
- Impedance loss due to charge line: $\hat{O}_{\mathrm{smf}} = \hat{n}$
- Quasiparticle loss: $\hat{O}_{\mathrm{smf}} = \sin(\hat{\phi}/2)$

Due to the presence of hybridizations between states with different P and hence different Φ_{eff}, the coupling of bath with fluxonium subsystem can mediate incoherent jumps between states with different P.

The Fermi's Golden Rule gives the transition rate between states $|\mu[P], P\rangle$ and $|\nu[P+1], P+1\rangle$ due to the channel α as:

$$\Gamma^{\alpha}_{\mu[P],P\to\nu[P+1],P+1} = \frac{1}{\hbar^2} \left| \left\langle \overline{\mu[P], P} \right| \hat{O}_{\mathrm{smf},\alpha} \left| \overline{\nu[P+1], P+1} \right\rangle \right|^2 S_{\alpha}(E_{\mu[P],P\to\nu[P+1],P+1}; T), \tag{56}$$

where $S_{\alpha}(E_{\nu[P+1],P+1} - E_{\mu[P],P}; T)$ is the noise spectral density of the bath α, evaluated at the energy difference $E_{\nu[P+1],P+1} - E_{\mu[P],P}$.

In the limit of weak hybridization, the expressions for the dressed eigenstates, the matrix element content of the decay rate is given by:

$$\begin{aligned}
&\left\langle \overline{\mu[P], P} \right| \hat{O}_{\mathrm{smf},\alpha} \left| \overline{\nu[P\pm 1], P\pm 1} \right\rangle \\
&= \underbrace{\langle \mu[P], P | \hat{O}_{\mathrm{smf},\alpha} | \nu[P\pm 1], P\pm 1\rangle}_{=0} \\
&+ E_{\mathrm{S}} \sum_{\alpha} \frac{T^{*}_{\alpha[P\pm1]\mu[P]} \langle \alpha[P\pm 1] | \hat{O}_{\mathrm{smf},\alpha} | \nu[P\pm 1]\rangle}{E_{\mu[P],P} - E_{\alpha[P\pm1],P\pm1}} \\
&+ E_{\mathrm{S}} \sum_{\alpha} \frac{T_{\nu[P\pm1]\alpha[P]} \langle \mu[P] | \hat{O}_{\mathrm{smf},\alpha} | \alpha[P]\rangle}{E_{\nu[P\pm1],P\pm1} - E_{\alpha[P],P}} \\
&+ \mathcal{O}(E_{\mathrm{S}}^2).
\end{aligned} \tag{57}$$

An important observation is that the decay rate is proportional to E_{S}^2, meaning that such jump rates depends on the array junction parameters and offset charges on the junction array islands. If we tune this E_{S} by adjusting junction array parameters (number, E_{J}, E_{C}, offset charges in each island), we can tune these jump rates.

Given that the observed jump rates are in level of Hz and smaller, which is much smaller than the typical depolarization rates between eigenstates in a fluxonium, it is a good approximation to consider that after the changes in the fluxoid numbers, the system quickly thermalizes within the states that has the same winding number. Define the thermal distribution function as:

$$p_{\mu[P],P}(T) = \frac{\exp\left(-\frac{E_{\mu[P],P}}{k_B T}\right)}{Z_P}, \tag{58}$$

with the partition function

$$Z_P = \sum_{\mu} \exp\left(-\frac{E_{\mu[P],P}}{k_B T}\right). \tag{59}$$

The total jump rate is then given by:

$$\Gamma_{P\to P\pm 1} = \sum_{\mu} p_{\mu[P],P} \times \sum_{\nu} \Gamma_{\mu[P],P\to\nu[P\pm 1],P\pm 1} \tag{60}$$

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